exact.

## Ⅱ. Tensor Product.

Def. Ⅱ.1.1. $M, N, P$ are $R$-modules. then $\exists$ $R$-module $T$ and $R$-bilinear $g: M\times N \to T$, has the Prop. property:

① $\forall$ $R$-module $P$. and $R$-bilinear $f: M\times N \to P$, $\exists! f': T \to P$. s.t. $f = f'\circ g$.

② If $(T, g)$, $(T', g')$ are the solution of ump, then $\exists!$ $j$ isomorphism, $T \to T'$. s.t. $j\circ g = g'$. then we will have $Bil_R(M, N; P) \cong Hom(M\otimes N; P)$.

And we can define $T = M\otimes N$.

$$\begin{array}{ccc} M\times N & \xrightarrow{g} & T := M\otimes N \\ & \searrow^{f} & \downarrow \exists! f' \text{ s.t. } f = f'\circ g \\ & & P \end{array}$$

Prop. Ⅱ.1.2. If $x_i \in M$, $y_i \in N$, $\sum x_i \otimes y_i = 0$ in $M\otimes N$. Then there exists $M' \subseteq M$. $N' \subseteq N$. finitely generated. s.t. $\sum x_i \otimes y_i = 0$ in $M'\otimes N'$.

Prop. Ⅱ.1.3. $M, N, P$ are $R$-modules then there exists isomorphism:

(i) $M\otimes N \cong N\otimes M$. (ii) $(M\otimes N)\otimes P \cong M\otimes N\otimes P \cong M\otimes(N\otimes P)$.

(iii). $(M\oplus N)\otimes P \cong (M\otimes P)\oplus(N\otimes P)$. (iv). $R\otimes M \cong M$.

Def II 1.4. $A, B$ rings. $N$ is $(A,B)$-module iff. $\forall a\in A, b\in B, x\in N$. $a(xb)=(ax)b$.

Prop. II 1.5. $A, B$ rings. $M$: $A$-module. $P$: $B$-module. $N$: $(A,B)$-module.

Then $(M\otimes_A N)\otimes_B P \cong M\otimes_A (N\otimes_B P)$.

Remark. $f: M\to M'$, $g: N\to N'$ are homomorphisms

Define $h: M\times N \to M'\times N' \to M'\otimes N'$

$(x,y)\to (f(x), g(y)) \to f(x)\otimes g(y)$.

introduces $R$-homomorphism: $f\otimes g: M\otimes N \to M'\otimes N'$.

$(f'\circ f)\otimes(g'\circ g) = (f'\otimes g')\circ(f\otimes g)$,

Def II 1.6. $f: A\to B$. $N$: $B$-module. then $N$ is $A$-module: $a\cdot x := f(a)\,x$ $(a\in A)$. (restriction of scalar)

Prop. II.1.7. $N$ is finitely generated $B$-module. $B$ is finitely $\sim A$-module. iff $N$ is finitely $\sim A$-module

Def II.1.8. $f: A\to B$. $M$ is $A$-module. then $B$ is $A$-module.

$M_B$ is $B$-module $:= B\otimes_A M$, $b(b'\otimes x) = bb'\otimes x$ (extension of scalar)

Prop. II.1.9. $M$ is fr-$A$-module. then $M_B$ is fr-$B$-module.

Lemma II.1.10. $M, N, P$ are $R$-module, then $\mathrm{Hom}(M\otimes N, P) \cong \mathrm{Hom}(M, \mathrm{Hom}(N,P))$.

Prop. II.1.11. Let $M' \xrightarrow{f} M \xrightarrow{g} M''\to 0$ ~~exact~~. $R$-module exact sequence

$N$: $R$-module. Then $M'\otimes N \xrightarrow{f\otimes 1} M\otimes N \xrightarrow{g\otimes 1} M''\otimes N \to 0$

Def II.1.12. $R$ ring, $T_N: M \mapsto M\otimes_A N$ on the category of $A$-modules and homomorphisms is exact. Then call $N$ is a ~~flat~~ $A$-module.

Prop. II.1.13. $N$ is $A$-module, then TFAE:

i) $N$ is flat;

ii) $0\to M'\to M\to M''\to 0$ exact $\Rightarrow$ $0\to M'\otimes N\to M\otimes N\to M''\otimes N\to 0$ exact.

iii). $f: M'\to M$ injective $\Rightarrow$ $f\otimes 1: M'\otimes N\to M\otimes N$ injective.

iv). $f: M'\to M$ injective, $M, M'$ finitely generated. then $f\otimes 1: M'\otimes N\to M\otimes N$ injecti

$f: A \to B$ ring-homomorphism. Then $M$ is $A$-flat $\Rightarrow M_B = B \otimes_A M$ is $B$-flat

for product on algebra

Def II.2.1. (i) $f: A \to B$ ring-homomorphism, if $\forall x, y \in B$. $a \in A$.

We have $(ax)y = x(ay) = a(xy)$. Then $B$ is $A$-algebra

(ii) $B, C$ are $A$-algebras, $h$ is $B \to C$ $A$-algebra homomorphism if

$B \xrightarrow{h} C$

$f \nwarrow A \nearrow g$.

(iii) $f: A \to B$. is finite $B$ is finite $A$-algebra if $B$ is fr-$A$-module.

(iv). $f: A \to B$ is finite type, $B$ is finitely-generated $A$-algebra

if $\exists\ x_1, \dots, x_n \in B$. s.t. $\forall x \in B$. $x \in f(A)[x_1, \dots, x_n]$.

(v). $A$ is finitely generated if $A$ as an $\mathbb{Z}$-algebra

Suppose $B, C$ are $A$-algebras. $f: A \to B$, $g: A \to C$. Then $D = B \otimes_A C$ is an $A$-algebra.

## III. Direct limit (without the language of category theory).

Def III.1.1. Let $I$ be a directed set. $A$ a ring. $(M_i)_{i \in I}$ a family of $A$-modules.

$\mu_{ij}: M_i \to M_j$ homomorphism. satisfying

(1) $\mu_{ii} = id_{M_i}$; (2) $\mu_{ik} = \mu_{jk} \circ \mu_{ij}$ $\forall i \le j \le k$.

$M = (M_i, \mu_{ij})$ ~~are~~ is called a direct system. over $I$. $l_i: M_i \to \bigoplus_{i \in I} M_i$ embedding.

Let $C = \bigoplus_{i \in I} M_i$. $D$ be the submodule of $C$ generated by elements: $l_i(x_i) - l_j \mu_{ij}(x_i)$ $(i \le j)$ $x_i \in M_i$

let $M = C/D$. $\mu: C \to M$ be the projection and let $\mu_i = \mu|_{M_i}: M_i \to M$

Then $(M, \mu_i)$ is called the direct limit of $M_i$, written $\varinjlim M_i$.

Prop III.1.2. (i). $\forall x \in \varinjlim M_i$, $\exists\ i \in I$. $x_i \in M_i$. s.t. $x = \mu_i(x_i)$.

(ii) If $\mu_i(x_i) = 0$. Then $\exists\ j \ge i$. s.t. $\mu_{ij}(x_i) = 0$.

pf: (skipped) $\dots \to M_i \xrightarrow{\mu_{ij}} M_j \xrightarrow{\mu_{jk}} M_k \to \dots$

Prop III.1.3. (the UMP of $\varinjlim M_i$). $N$: $R$-module $\alpha_i: M_i \to N$ homomorphism.

$\forall i \le j$. $\alpha_i = \alpha_j \circ \mu_{ij}$ then $\exists!\ \alpha: \varinjlim M_i \to N$, $\alpha_i = \alpha \circ \mu_i$ $\forall i \in I$

pf: Consider the diagram:

